

Analysis

1 Order Parameters

To characterize the orientational organization of elongated cells, the following global order parameters are recorded.

The global nematic order is defined as

$$Q_{E,\text{elongated}}(t) = \frac{1}{N_E(t)} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(2\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(2\varphi_i(t)) \right]^2}. \quad (1)$$

The global polar order is defined as

$$P_{E,\text{elongated}}(t) = \frac{1}{N_E(t)} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(\varphi_i(t)) \right]^2}. \quad (2)$$

We also calculate the parameters normalized by the total number of cells:

$$Q_{E,\text{total}}(t) = \frac{1}{N} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(2\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(2\varphi_i(t)) \right]^2}, \quad (3)$$

and

$$P_{E,\text{total}}(t) = \frac{1}{N} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(\varphi_i(t)) \right]^2}. \quad (4)$$

The phenotypic composition is characterized by

$$f_E(t) = \frac{N_E(t)}{N}. \quad (5)$$

These quantities satisfy

$$P_{E,\text{total}}(t) = f_E(t) P_{E,\text{elongated}}(t), \quad Q_{E,\text{total}}(t) = f_E(t) Q_{E,\text{elongated}}(t). \quad (6)$$

When $N_E(t) = 0$, the order parameters normalized by the elongated population are set to zero.

1.1 Time evolution of order parameters

We first analyze the time evolution of the order parameters.

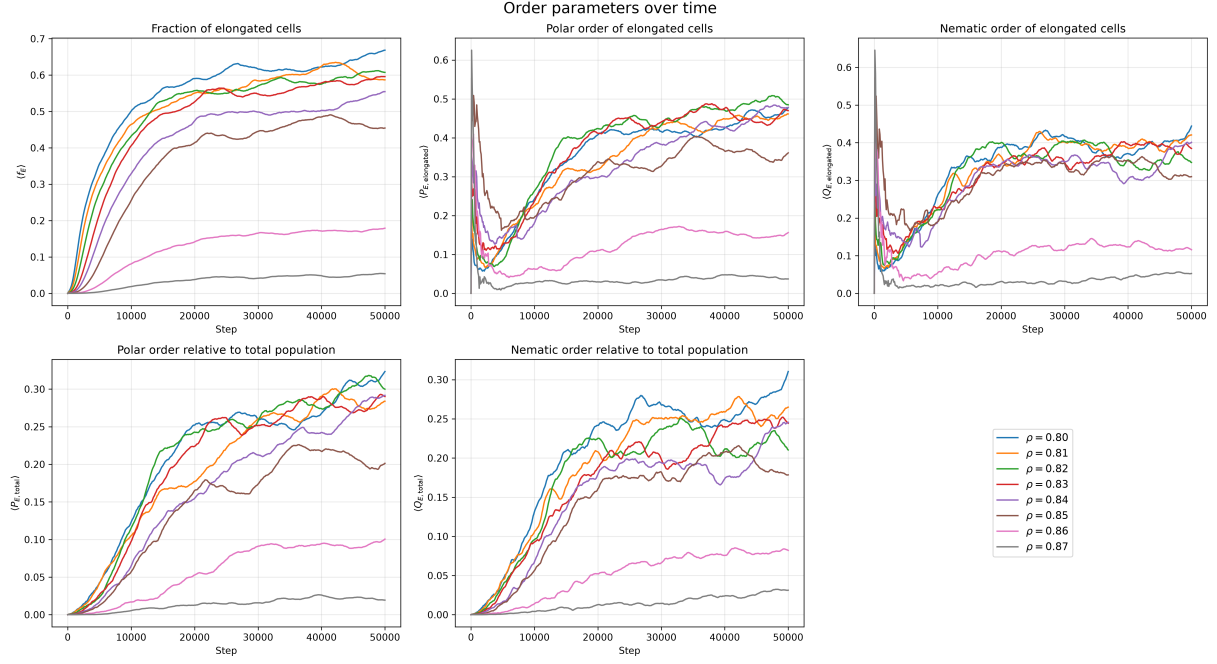


Figure 1: Time evolution of the elongated fraction and orientational order parameters for the different densities. Curves are averaged over independent realizations.

f_E increases progressively from the initially round configuration. For lower densities, a substantial fraction of the population becomes elongated, while the elongated fraction is strongly suppressed at the highest densities.

$P_{E,\text{total}}$ and $Q_{E,\text{total}}$ show the same suppression at high density, indicating that collective orientational order becomes macroscopically small when elongation is inhibited.

$P_{E,\text{elongated}}$ and $Q_{E,\text{elongated}}$ remain finite whenever an elongated subpopulation is present. At very early times, these quantities should be interpreted with caution because N_E is still very small.

1.2 Steady state

To obtain a more stable estimate of the order parameters and compare their dependence on density, we average each observable over the final 10% of each realization. We first perform the temporal average independently for each seed and then average over realizations.

At this stage, the final 10% is used as a late-time averaging window rather than assuming that all realizations have reached a fully stationary state. Longer simulations will be used to assess convergence.

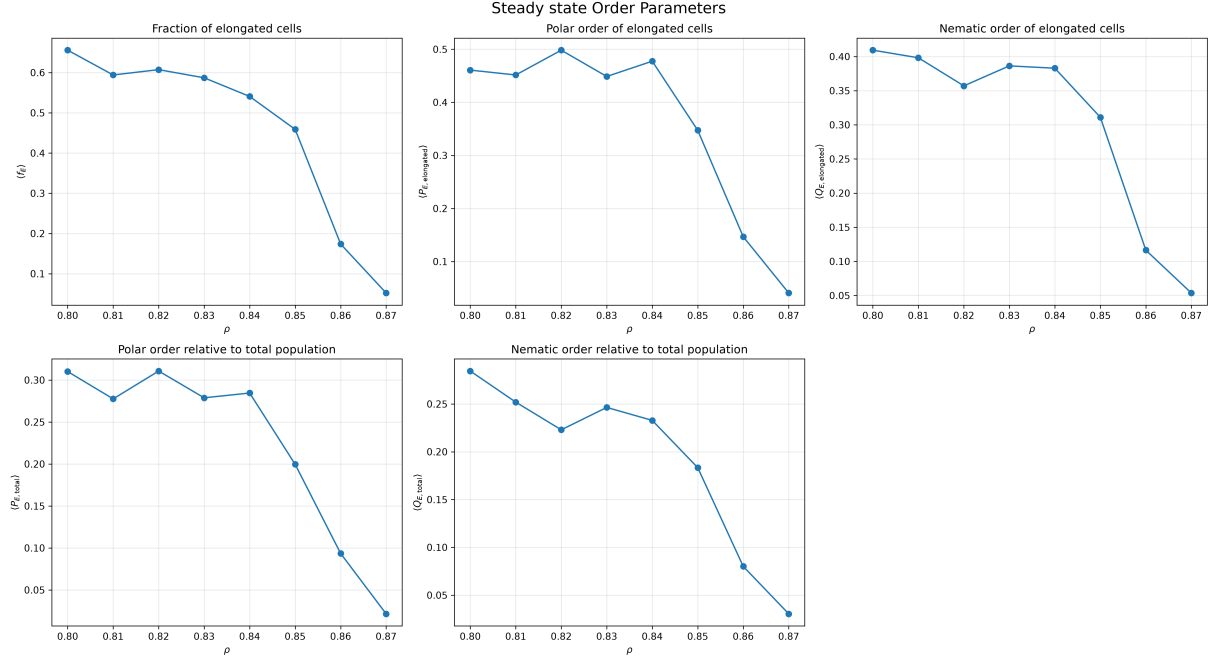


Figure 2: Late-time order parameters as a function of density. Each point is obtained by first averaging over the final 10% of each realization and then averaging over realizations.

The late-time f_E remains relatively large at lower densities and decreases sharply between $\rho = 0.85$ and $\rho = 0.86$. A similar decrease is observed for $P_{E, \text{total}}$ and $Q_{E, \text{total}}$.

Among the quantities considered here, f_E provides the most direct measure of the morphological state of the system and will therefore be used as the primary observable to characterize the transition.

1.3 Susceptibility

To quantify the fluctuations of the elongated fraction, we define its susceptibility as

$$\chi_{f_E} = N_{\text{actual}} \left[\langle f_E^2 \rangle - \langle f_E \rangle^2 \right]. \quad (7)$$

The temporal moments are first evaluated within the late-time window for each realization and are subsequently averaged over seeds.

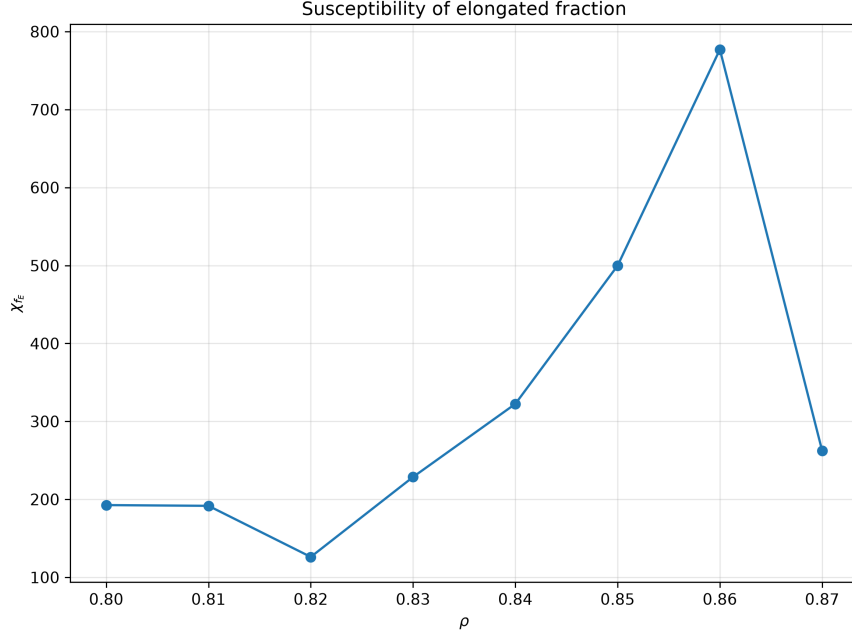


Figure 3: Susceptibility of the elongated fraction as a function of density for $N = 10000$.

For $N = 10000$, the susceptibility displays a pronounced maximum at $\rho = 0.86$, coinciding with the region where the late-time elongated fraction decreases sharply. We therefore use the susceptibility maximum as a reference point to inspect the fluctuations and the distribution of f_E near the transition.

1.4 Elongated-fraction distributions near the susceptibility peak

To investigate the origin of the enhanced fluctuations, we examine the distribution of the late-time averaged elongated fraction, $\langle f_E \rangle_t$, across independent realizations. We consider the density corresponding to the susceptibility maximum together with the immediately neighboring sampled densities.

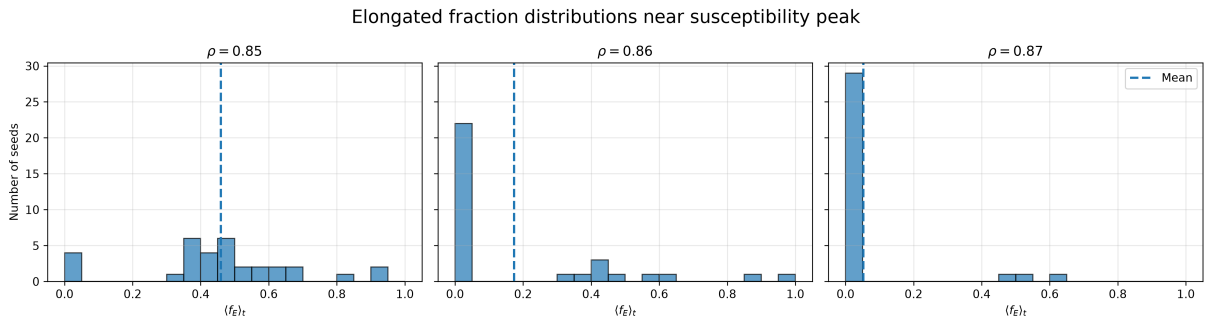


Figure 4: Distribution across independent realizations of the late-time averaged elongated fraction near the susceptibility maximum.

At $\rho = 0.85$, most realizations retain a substantial elongated population, although realizations with very small f_E are already present. At $\rho = 0.86$, realizations with $f_E \approx 0$ coexist with others retaining a finite elongated fraction. At $\rho = 0.87$, most realizations belong to the low- f_E branch, while a small number remain strongly elongated.

1.5 Individual realizations near the susceptibility peak

To determine whether the broad distribution near the susceptibility maximum originates from distinct dynamical behaviors, we inspect the time evolution of f_E for each independent realization at $\rho = 0.86$.



Figure 5: Time evolution of the elongated fraction for individual realizations at $\rho = 0.86$.

The individual trajectories show a strong separation between realizations. Several remain close to the fully round state, whereas others develop and retain a substantial elongated population.

2 Cluster analysis

To characterize the spatial organization of the system, clusters are identified independently for round and elongated cells. Two cells are considered connected when they belong to the same phenotype and interact significantly according to the same overlap criterion used in the dynamics. Connectivity is transitive, and clusters are reconstructed independently for each saved configuration. Isolated cells are therefore retained as clusters of size one.

For each phenotype $\alpha \in \{R, E\}$, we denote by N_α the number of cells belonging to that phenotype and by $S_{1,\alpha}$ and $S_{2,\alpha}$ the sizes of the largest and second-largest clusters, respectively.

The size of the largest cluster can be normalized either by the total number of cells,

$$\frac{S_{1,\alpha}}{N},$$

or by the number of cells belonging to the corresponding phenotype,

$$\frac{S_{1,\alpha}}{N_\alpha}.$$

These quantities provide complementary information. The first measures the fraction of the entire system contained in the largest cluster of phenotype α , whereas the second measures how strongly the population of that phenotype is organized into a single connected structure.

They are related through

$$\frac{S_{1,\alpha}}{N} = \frac{N_\alpha}{N} \frac{S_{1,\alpha}}{N_\alpha}.$$

Thus, changes in the macroscopic size of a cluster may arise either from changes in the abundance of a phenotype or from changes in its internal spatial organization.

To quantify the relative dominance of the largest cluster over the second-largest one, we define

$$\psi_\alpha = \frac{S_{1,\alpha} - S_{2,\alpha}}{S_{1,\alpha} + S_{2,\alpha}}.$$

Values of ψ_α close to one indicate a strongly dominant largest cluster, whereas smaller values indicate that the two largest structures have comparable sizes.

In addition to the largest clusters, we analyze the fragmentation of each population through the number of clusters, the fraction of isolated cells, and the distribution of finite-cluster sizes. For elongated clusters, their internal polar and nematic order are also available, together with observables describing the collective motion of each cluster.

2.1 Time evolution of cluster structure

We first study how the cluster structure develops from the initially round configuration.

For each phenotype, the fraction of the total system contained in its largest cluster can be written as

$$\frac{S_{1,\alpha}}{N} = \frac{N_\alpha}{N} \frac{S_{1,\alpha}}{N_\alpha}.$$

This decomposition allows us to distinguish two contributions to the growth or disappearance of a macroscopic cluster: changes in the abundance of the corresponding phenotype, N_α/N , and changes in the connectivity of that population, measured by $S_{1,\alpha}/N_\alpha$.

We therefore examine the time evolution of these quantities separately for round and elongated cells.

We start plotting $\frac{S_{1,\alpha}}{N}$:

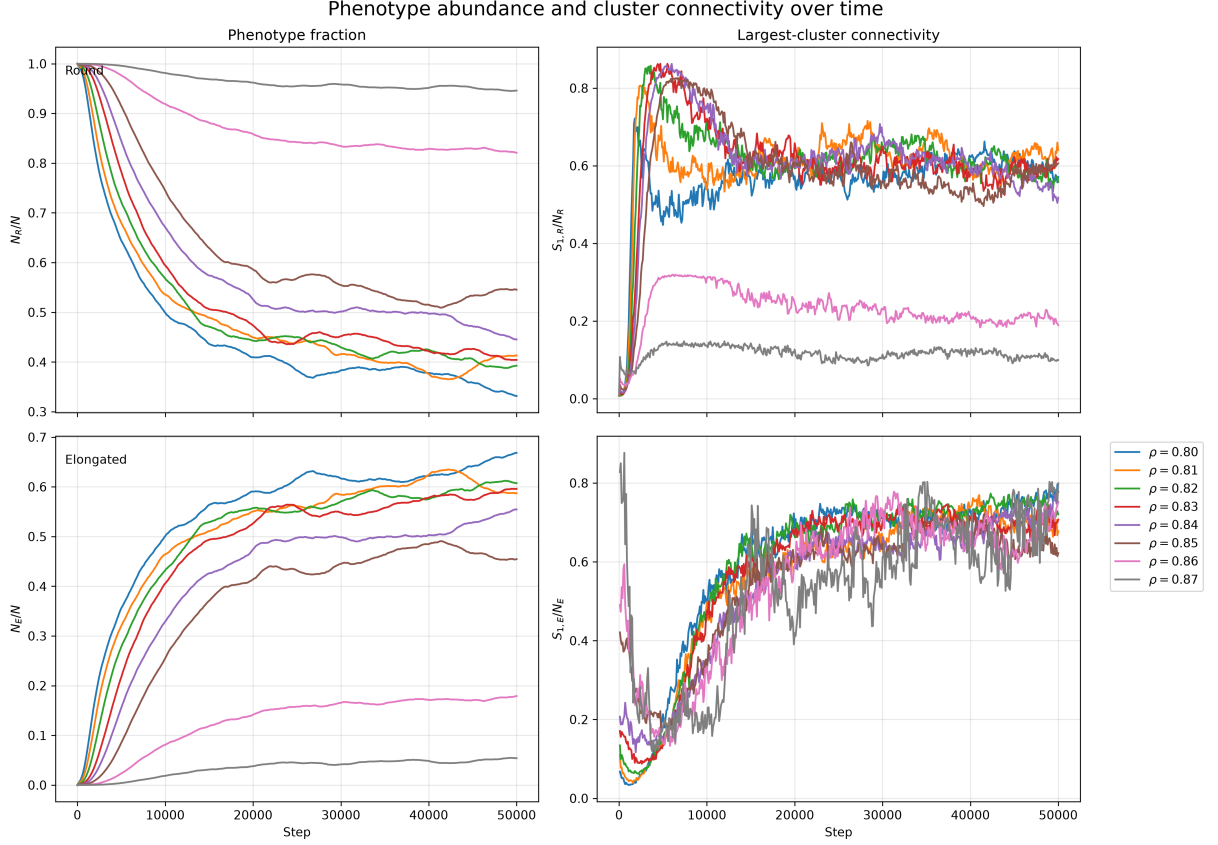


For elongated cells, the fraction of the total system contained in the largest cluster decreases at the highest densities. This behavior is expected to be closely related to the suppression of the elongated population observed previously through f_E .

The behavior of round cells is qualitatively different. Although $S_{1,R}/N$ initially increases with density (compare $\rho = 0.80$ and $\rho = 0.85$), it decreases again for the highest densities, particularly around $\rho = 0.86$ and $\rho = 0.87$. This is not simply explained by the phenotypic composition, since these densities correspond to systems dominated by round cells.

This suggests that a high fraction of round cells does not necessarily imply that they form a single macroscopic connected structure. In particular, the suppression of elongation at high density may result from geometric constraints even in the absence of a dominant interacting cluster of round cells.

To separate changes in phenotypic abundance from changes in connectivity, we now examine N_α/N and $S_{1,\alpha}/N_\alpha$ independently.



The decomposition shows that phenotypic abundance and cluster connectivity behave quite differently.

For round cells, N_R/N increases strongly with density, but at the highest densities $S_{1,R}/N_R$ decreases. So even though most cells are round, they do not form a large interacting cluster according to our current definition.

My guess is that elongation is still blocked by crowding and spatial constraints, even without a giant interaction-based cluster. If we used a more geometric cluster definition, for example connecting cells within a distance R , we might actually recover a giant cluster in this regime.

For elongated cells, most of the density dependence comes from N_E/N . When a substantial elongated population exists, a large fraction of it usually belongs to the largest elongated cluster.

So the decrease of $S_{1,E}/N$ at high density seems to be mainly due to the disappearance of elongated cells, rather than to strong fragmentation of the elongated population.

At very early times, and also at the highest densities, $S_{1,E}/N_E$ should be interpreted carefully because N_E is very small.

2.2 Steady state cluster structure

2.2.1 Largest cluster size

To compare the cluster structure between densities, we average the cluster observables over the final 10% of each realization. As in the order-parameter analysis, this window is used to

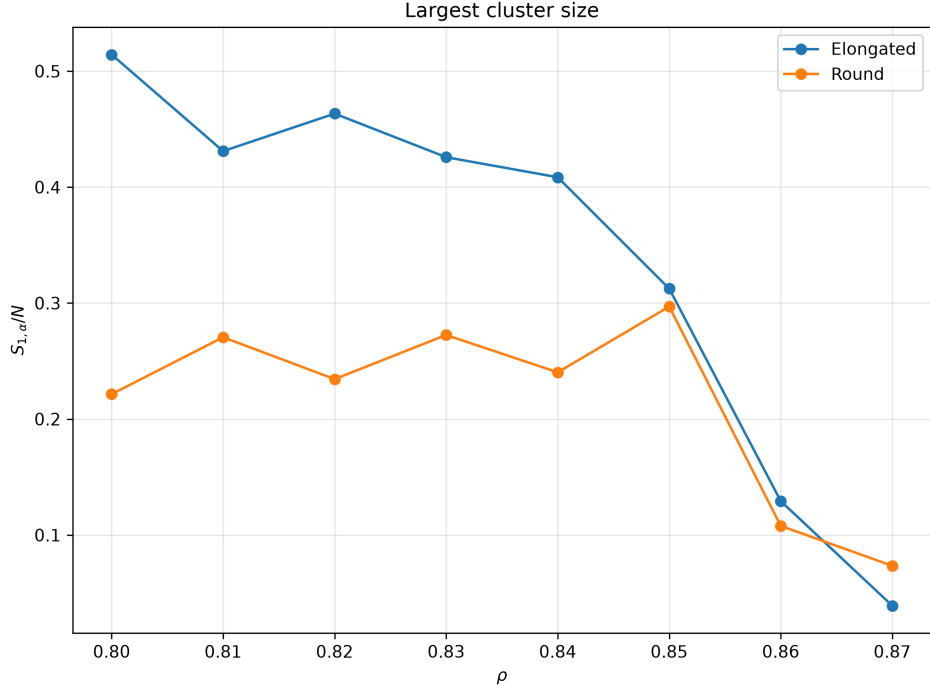
obtain more stable late-time estimates without assuming that every realization has reached a fully stationary state.

We first average over time independently for each seed, and then average over realizations.

We first consider the fraction of the whole system contained in the largest cluster of each phenotype,

$$\frac{S_{1,\alpha}}{N}.$$

This gives a direct measure of the macroscopic importance of the largest round and elongated structures as density is varied.

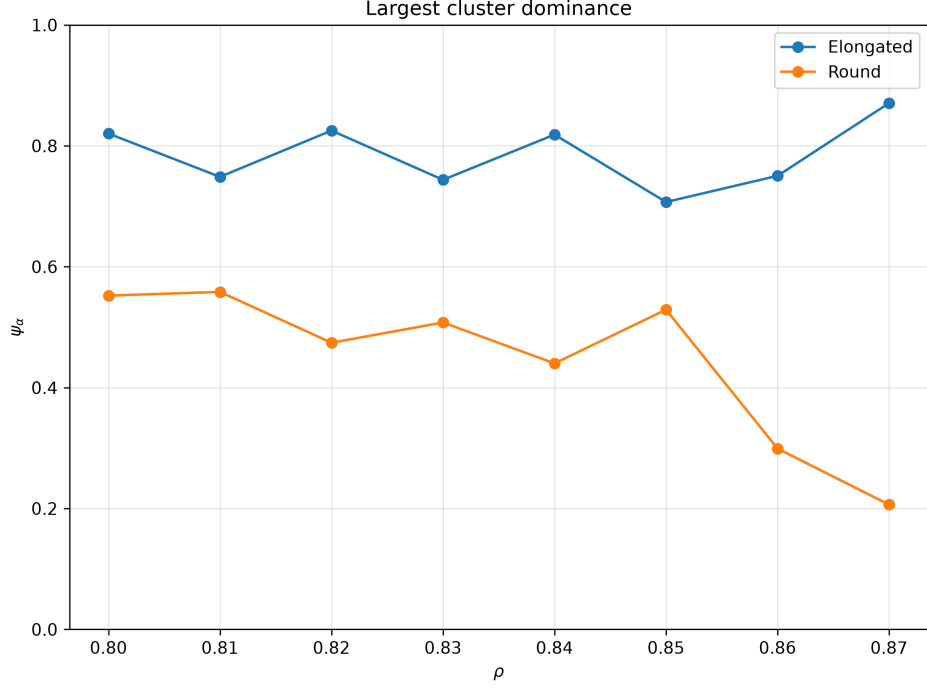


2.2.2 Largest-cluster dominance

The size of the largest cluster does not tell us whether it is clearly dominant over the remaining structures. We therefore also consider

$$\psi_\alpha = \frac{S_{1,\alpha} - S_{2,\alpha}}{S_{1,\alpha} + S_{2,\alpha}}.$$

A value close to one corresponds to a clearly dominant largest cluster, while smaller values indicate that the two largest clusters have more comparable sizes.



For elongated cells, ψ_E remains relatively large over the whole density range, suggesting that the largest elongated cluster is usually clearly dominant over the second-largest one.

For round cells, ψ_R decreases strongly at $\rho = 0.86$ and $\rho = 0.87$. Together with the decrease of $S_{1,R}/N_R$, this suggests that the round population becomes less dominated by a single connected structure at high density.

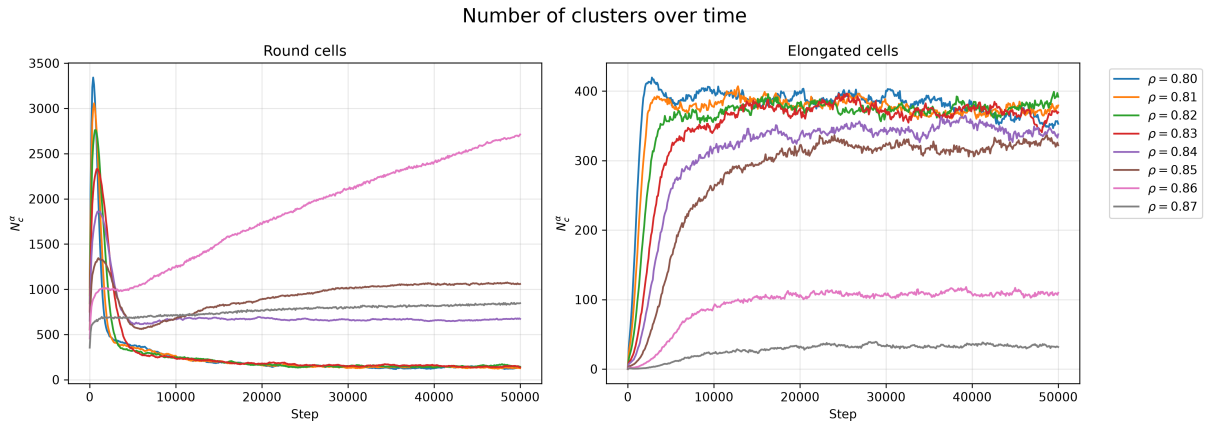
However, ψ_R alone does not tell us whether this corresponds to a few comparable clusters or to fragmentation into many smaller clusters. To check this, we next look at the number of clusters.

2.3 Fragmentation

The previous results suggest that the round population becomes less dominated by a single connected structure at high density. We now ask whether this corresponds to a fragmentation into a larger number of clusters.

2.3.1 Time evolution of the number of clusters

We first look at the total number of clusters, N_c^α , for each phenotype as a function of time.

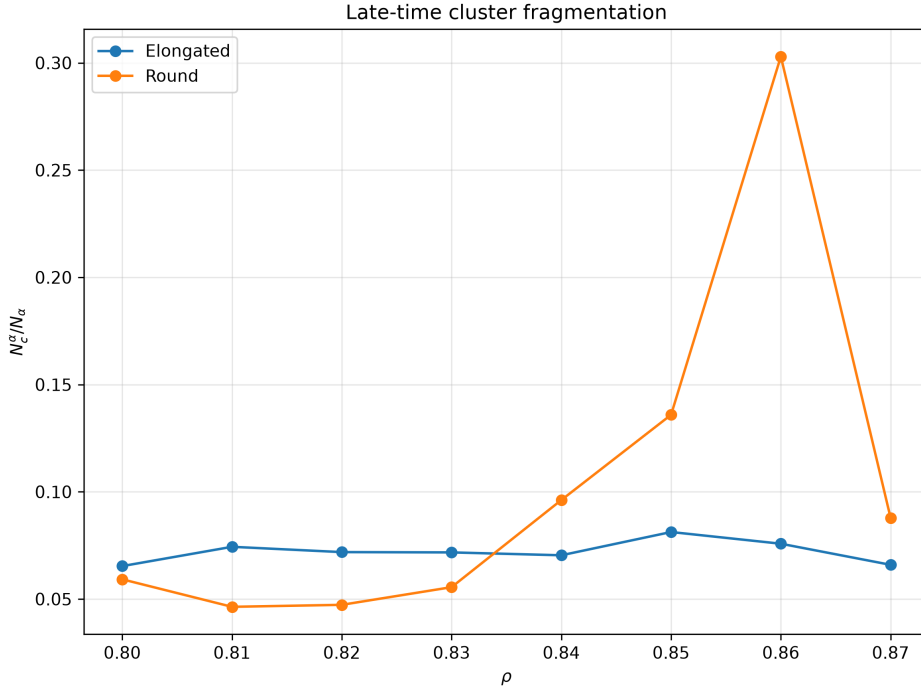


The number of clusters shows a strong change in the organization of the round population.

At lower densities, the number of round clusters initially increases but then decreases strongly as the system evolves. Around $\rho = 0.85$ and especially $\rho = 0.86$, many more round clusters remain at late times. In particular, at $\rho = 0.86$ the number of round clusters continues increasing during most of the simulation, which is consistent with the low values of $S_{1,R}/N_R$ and ψ_R found before.

Interestingly, $\rho = 0.87$ contains fewer round clusters than $\rho = 0.86$, even though the round fraction is larger. This suggests that the strongest fragmentation may occur around the transition region rather than simply increasing with density.

For elongated cells, the number of clusters decreases strongly at high density. However, this is partly expected because the total elongated population also becomes very small, so the raw number of clusters alone is not enough to quantify their fragmentation.



The fragmentation of the round population changes strongly with density.

At lower densities, where a large elongated population develops, the round cells form relatively few clusters and a large fraction of them belongs to the largest one. One possibility is that the motion and organization of the elongated cells pushes the remaining round cells together, helping them form a larger connected structure.

Around $\rho = 0.86$, the situation changes strongly. The normalized number of round clusters has a clear maximum, while $S_{1,R}/N_R$ and ψ_R are small. My current interpretation is that this may be a geometric blocking regime: there is already little space available for elongation, but the round cells are still not dense enough to become strongly connected through our interaction criterion.

This is also the density where different realizations behave very differently. Some remain almost completely round, while others develop a substantial elongated population, so this seems to be a region where several different structures can appear.

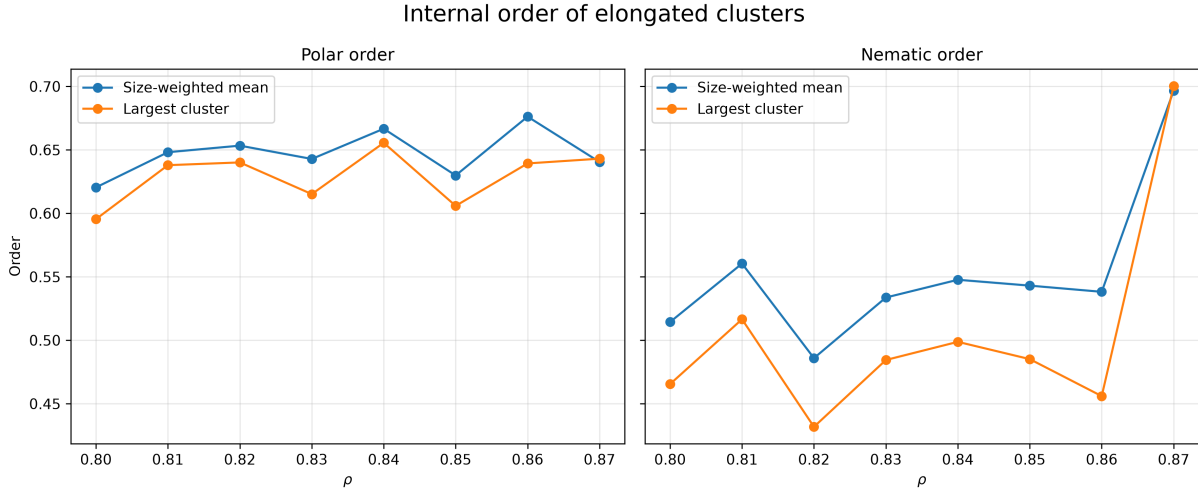
At $\rho = 0.87$, almost all cells remain round and the number of clusters decreases again. A possible explanation is that the higher density increases the number of contacts between round cells, joining the small components into larger ones. However, there is still no clearly dominant round cluster, as seen from the low values of $S_{1,R}/N_R$ and ψ_R .

2.4 Internal order of elongated clusters

The global polar and nematic order parameters studied before do not tell us whether the elongated clusters are internally ordered. A low global order could arise either because the clusters themselves are disordered or because different ordered clusters point in different directions.

We therefore look at the polar and nematic order inside elongated clusters. Single-cell clusters are excluded, since their polar and nematic order are trivially equal to one.

We compare the size-weighted mean order of the elongated clusters with the order of the largest elongated cluster.



The internal order of elongated clusters changes relatively little with density.

The polar order remains around 0.6 – 0.7 over most of the density range, both for the size-weighted average and for the largest cluster. This suggests that, whenever elongated clusters form, they remain internally quite ordered.

The nematic order is somewhat lower, and the largest cluster is generally slightly less nematically ordered than the size-weighted average.

Overall, the disappearance of the elongated state at high density does not seem to be caused by a progressive loss of internal cluster order. Instead, the main change appears to be the disappearance of the elongated population itself.

The values at $\rho = 0.87$ should be interpreted carefully, since very few elongated cells remain and the corresponding clusters are small.

2.5 Cluster motion

Finally, we look at how the motion of elongated clusters depends on their size.

For each cluster $\mathcal{C}$, we distinguish between its collective speed,

$$V_{\mathcal{C}} = \left| \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \mathbf{v}_i \right|,$$

and the mean speed of its cells,

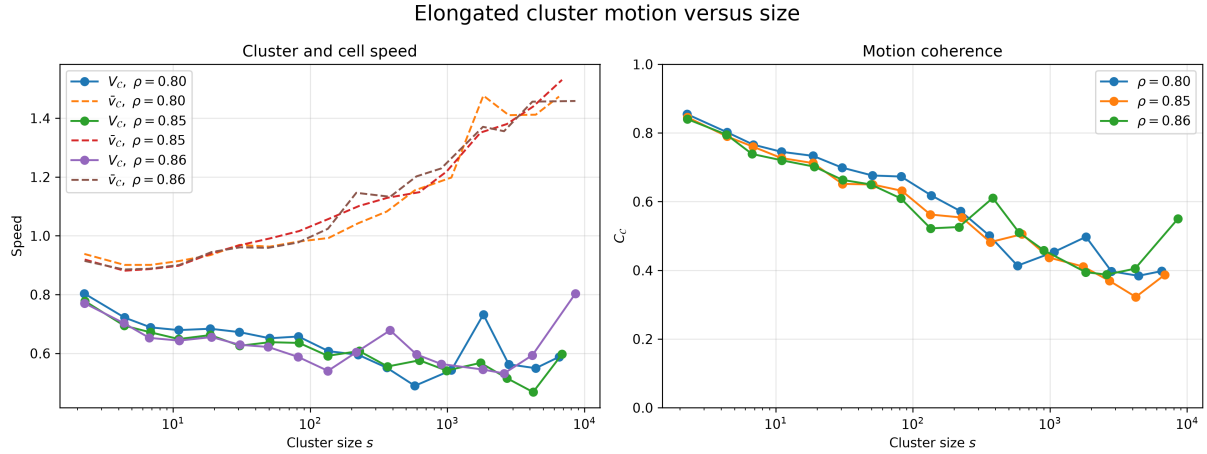
$$\bar{v}_{\mathcal{C}} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} |\mathbf{v}_i|.$$

Their ratio,

$$C_{\mathcal{C}} = \frac{V_{\mathcal{C}}}{\bar{v}_{\mathcal{C}}},$$

gives a simple measure of motion coherence. Values close to one indicate that most of the cellular motion contributes to translation of the cluster, while smaller values indicate stronger cancellation of the individual cell velocities.

We study how these quantities change with cluster size.



The cluster motion shows a clear dependence on cluster size.

For small elongated clusters, the collective speed V_C is relatively close to the mean cell speed $\bar{v}_C$, giving a motion coherence C_C close to one.

As the cluster size increases, the mean cell speed increases, while the collective speed remains smaller and changes much less. As a result, C_C decreases systematically with cluster size.

This suggests that cells in large elongated clusters remain active, but an increasing part of their motion is internal and cancels when calculating the collective velocity. This could be related to curvature or internal variations of the direction of motion.

Interestingly, this behavior is quite similar for the different densities considered here, suggesting that cluster size may be more important than density for the coherence of cluster motion.

The largest-size bins should be interpreted carefully, since they contain fewer clusters.